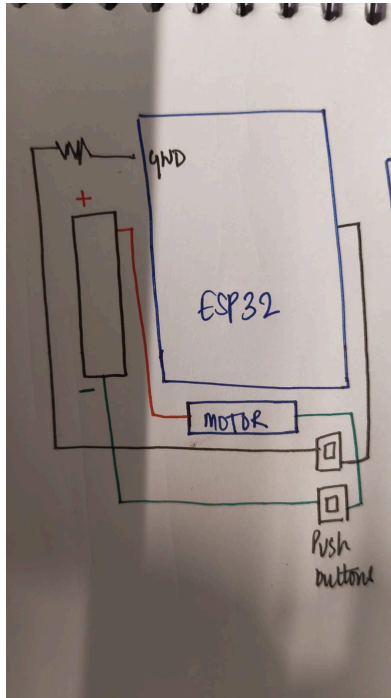


Electronics Documentation

1. Electronics Block Diagram

Show signal and power flow



2. Components Table

Component	Quantity	Purpose	Justification
ESP32	1	To serve as main microcontroller	Multiple functions like bluetooth, motors and PWM required
Coreless brushed Motor	1	To achieve spin	Coreless motors spin faster than stepper motor for the kind of motion required.
Push buttons	2	To control motor and serve as	The buttons are a part of 2 different circuits. Since

		<i>user input for accuracy calculation</i>	<i>connecting a battery to the esp32 is a safety hazard, the two buttons collectively collect user input: one allows the user to actually perform the input act (spin the cat), the other inputs the data required for calculations into the esp32. (time held, tally) this was possible only by merging these buttons.</i>
<i>LED</i>	<i>1</i>	<i>To aid in user experience</i>	<i>Programmed to output a strobe pattern, the LED aids in simulating a club/rave experience for the user, staying true to the original cat rave. It is meant to aid users into indulging in the atmosphere and semiotically understand the game subconsciously.</i>

Justification is important

Contents of the Folder

1. Electronic Details Document
2. Circuit Diagram

 **Guidelines** Ensure diagrams are neat and readable Can be hand-drawn or software-generated Should match actual implementation Justification of components is mandatory

Fabrication and Integration Process

1. Material Choice Justification :

We chose recycled materials from packaging to maintaining structural integrity for a spinning motor.

2. Fabrication Steps :

A documented "Build-log" (e.g., "1. Laser cutting frame," "2. Heat-staking threaded inserts").

- 1) Prepared the frame to fit the breadboard*
- 2) Attached it together with masking tape to test out resistance to torque*
- 3) Set up shade for strobe light*
- 4) Used adhesive to fix frame after final testing*
- 5) 3D printed button to fit over the push buttons simultaneously*
- 6) Experimented with adhesives to attach the visualizer to the motor*

Contents of the folder

- 1. Drawings & Sketches: Initial ideas, Concept sketches, Iterations*
- 2. CAD / Simulations: 3D models (if applicable), Simulation results (if any)*
- 3. Fabrication & Integration Process Document*

Guidelines

- 1. Show evolution of design (not just final result)*
- 2. Explain why materials were chosen*

Software Documentation

1. Software Overview

- What does the software do?*
- Platforms used: MicroPython, MIT App Inventor*

The app does not have much purpose on the functional end, apart from connecting to the esp32 via bluetooth, marking the start of the timer which controls the LED's inception and the beginning of the tally recording. The app merely only interacts with the user and aids in creating a lighthearted atmosphere - the most important one playing the

audio itself. Micropython is the logical centre of the system. It keeps a track of user tallies, compares it against the control dataset, and calculates the final accuracy score of the user. Micropython also controls the pattern of the LED.

2. System Logic

Explain flow in simple steps

- 1. User connects to bluetooth app*
- 2. User presses start button and places one hand over the physical button of the stage*
- 3. When the song begins, there is a time window at every 'oia' where the code will add to a tally if the button is clicked accurately.*
- 4. Along with a tally, the amount of time the button has been pressed down for will also be recorded.*
- 5. The "pressed-down" values are compared against a control dataset, and each tally is assigned an accuracy value, as determined by the difference of the control and user input*
- 6. At the end of the song, the cumulative accuracy value score will appear.*
- 7. The score is only calculated when the number of user input tallies is equal to the control tallies.*

3. Flowchart

<https://miro.com/app/board/uXjVHbwkr9g=/>

4. App Flow (If MIT App Inventor)

<https://miro.com/app/board/uXjVHbwkr9g=/>